

Jiacheng Xu

 Robot Learning Researcher

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 KTH Royal Institute of Technology

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M.Sc. student in Applied Mathematics (Scientific Computing) at KTH Royal Institute of Technology. My research focuses on robot learning and physics-based priors for manipulation. I am interested in scalable learning systems for embodied agents.

Education

Sep 2023	Mathematics, KTH Royal Institute of Technology & TU Berlin
June 2026	Master in Scientific Computing Courses: Machine Learning; GPU Programming; Computer Vision; Numerical Methods for PDEs; Large-Scale Matrix Computations; Model Reduction
Sep 2022	Electronic Eng., University of Bath
June 2023	Master in Robotics Courses: Reinforcement Learning; ROS
Sep 2018	Mechatronic Eng., Qingdao Institute of Technology
June 2022	Bachelor in Mechanical Engineering

Selected Projects

Robot Sim-to-Real Suite

Research Engineer

Oct 2025 – Current

- › Built a lightweight sim-to-real evaluation framework using SAPIEN for policy rollout across simulation and real robot hardware.
- › Implemented teleoperation pipeline for humanoid robot to collect demonstrations for fine-tuning GR00T robot foundation models.
- › Deployed a Diffusion Policy inference web service and integrated ROS control for real-time dexterous-hand execution.

Gaussian Splatting for Language-Driven Manipulation (3DGS)

Master's Thesis

Oct 2024 – Sep 2025

- › Built semantic-aware 3D Gaussian Splatting by distilling 2D vision models into per-Gaussian features.
- › Structured LLM outputs into executable optimization objectives for downstream robot planning.
- › Enabled instruction-conditioned 3D scene editing and post-state reconstruction (e.g., relocating objects to satisfy task constraints).

Multibody Dynamics Simulator

Personal Project

Oct 2025 - Nov 2025

- › Constructed a physics engine with pure Rust, simulating 6-DOF rigid chains under gravity.
- › Implemented Lie Algebra-based spatial vector calculus: Applied CRBA (Composite Rigid Body Algorithm) and RNE (Recursive Newton-Euler) for force/torque solving.
- › Wrote an RK4 (Runge-Kutta 4) numerical integrator to ensure stability and implemented a real-time 3D visualization.

Bachelor Awards

- › National Smart Car Competition (2nd Prize)

- › National Electronics Design Contest (Provincial 1st)
- › National Mathematical Modeling Contest (2nd Prize)

Work Experience

Oct 2024	Backend Developer @ Alfer Tech (Startup)
May 2025	› Built a PDF QA system with Retrieval-Augmented Generation (RAG) Semantic Search; › Integrated with PDF.js navigation, capable for hundreds of documents.

Skills

Programming	Python (PyTorch), C++, Rust, CUDA
Simulation	MuJoCo, IsaacLab, SAPIEN, Genesis
App Dev	FastAPI/Flask, React
A-Z Languages	Chinese (native), English (fluent), Swedish (beginner)